

Combinatorics & Fast Power

Factorials, nCr , Pascal's Triangle, Fast Exponentiation, floor/ceil/mod · Mentor-Led Lab

Developed by TalenciaGlobal

Difficulty ★★★ Intermediate

Problem

Given a handful of inputs, compute several combinatorial quantities, a modular power using fast exponentiation, and the floor, ceiling and remainder of a division.

What to Compute

- $n!$ (the factorial of n).
- nCr , the binomial coefficient $C(n, r)$.
- The n -th row of Pascal's triangle (its $n + 1$ coefficients).
- base raised to exp, modulo mod, computed with fast (binary) exponentiation.
- $\text{floor}(p / q)$, $\text{ceil}(p / q)$ using the $(p + q - 1) / q$ trick, and $p \bmod q$.

Input / Output Format

Input: Seven integers: n r base exp mod p q .

Output: Seven labelled lines: Factorial, nCr , Pascal Row, Fast Power, Floor, Ceil, Mod.

Examples

Example 1

Input: $n=5$, $r=2$, base=3, exp=13, mod=7, $p=17$, $q=5$

Output:

```
Factorial = 120
nCr = 10
Pascal Row = 1 5 10 10 5 1
Fast Power = 3
Floor = 3
Ceil = 4
Mod = 2
```

Explanation: $5! = 120$; $C(5,2) = 10$; row 5 is 1 5 10 10 5 1; $3^{13} \bmod 7 = 3$ (because $3^6 \bmod 7 = 1$); $\text{floor}(17/5) = 3$; $\text{ceil}(17/5) = 4$; $17 \bmod 5 = 2$.

Example 2

Input: $n=6$, $r=3$, base=2, exp=10, mod=1000, $p=10$, $q=3$

Output:

```
Factorial = 720
nCr = 20
Pascal Row = 1 6 15 20 15 6 1
Fast Power = 24
Floor = 3
Ceil = 4
Mod = 1
```

Explanation: $6! = 720$; $C(6,3) = 20$; row 6 is 1 6 15 20 15 6 1; $2^{10} \bmod 1000 = 24$; $\text{floor}(10/3) = 3$; $\text{ceil}(10/3) = 4$; $10 \bmod 3 = 1$.

Constraints

- $0 \leq r \leq n \leq 20$ (so $n!$ and $C(n, r)$ fit in a 64-bit integer)
- $0 \leq \text{base}, \text{exp} \leq 10^9$ and $1 \leq \text{mod} \leq 10^9$
- $0 \leq p$ and $1 \leq q$

Approach

Factorial is the simple product $1 * 2 * \dots * n$.

Compute nCr incrementally as $\text{res} = \text{res} * (n - r + i) / i$ for $i = 1..r$; every partial product is a whole number, so this avoids building huge factorials.

Build Pascal's row with the same idea: start at 1 and multiply by $(n - k) / (k + 1)$ to step from $C(n, k)$ to $C(n, k + 1)$.

Fast exponentiation squares the base and halves the exponent, reducing modulo mod at each step; it uses $O(\log \text{exp})$ multiplications instead of exp of them.

For the ceiling of a non-negative division, $(p + q - 1) / q$ gives the rounded-up quotient using only integers.

Sample Run**Sample Input**

```
5 2 3 13 7 17 5
```

Sample Output

```
Factorial = 120
nCr = 10
Pascal Row = 1 5 10 10 5 1
Fast Power = 3
Floor = 3
Ceil = 4
Mod = 2
```